

KYSHTOVKA, Russian Federation

WMO: 294050

Lat: 56.6011N

Lon: 76.5613E

Elev: 116

StdP: 99.94

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.1	-34.0	-40.2	0.1	-36.9	-37.2	0.1	-33.9	9.1	-11.5	7.9	-10.3	1.5	0	0.647

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	28.9	19.2	27.4	18.5	25.7	17.8	21.0	26.5	19.9	25.4	18.9	23.9	3.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	14.0	23.6	17.9	13.0	22.6	16.9	12.2	21.5	61.5	26.5	57.6	25.5	54.3	23.9	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	7.9	6.7	-40.5	31.6	4.1	1.7	-43.4	32.8	-45.8	33.8	-48.0	34.8	-51.0	36.0		
(5)			-40.6	22.9	3.8	1.9	-43.4	24.3	-45.6	25.4	-47.8	26.5	-50.6	27.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-19.4	-15.8	-7.1	2.6	10.4	18.3	15.3	9.4	2.2	-8.3	-15.7	(6)	
(7)		DBStd	14.39	8.35	7.50	6.03	5.80	5.48	4.33	3.40	3.71	4.31	5.04	8.11	8.52	(7)
(8)		HDD10.0	4121	912	723	530	229	64	5	0	3	62	246	549	798	(8)
(9)		HDD18.3	6496	1171	956	788	471	249	84	43	109	268	501	799	1056	(9)
(10)		CDD10.0	754	0	0	0	8	78	198	257	166	44	3	0	0	(10)
(11)		CDD18.3	89	0	0	0	0	5	27	42	14	1	0	0	0	(11)
(12)		CDH23.3	989	0	0	0	1	111	317	384	157	18	0	0	0	(12)
(13)		CDH26.7	210	0	0	0	0	24	69	85	29	3	0	0	0	(13)
(14)	Wind	WSAvg	3.1	2.7	3.0	3.4	3.6	3.7	2.9	2.7	3.0	3.2	3.3	3.0	(14)	
(15)	Precipitation	PrecAvg	428	25	19	19	21	38	53	58	36	37	35	30	(15)	
(16)		PrecMax	598	51	35	52	44	92	137	107	122	79	71	65	63	(16)
(17)		PrecMin	244	1	1	1	0	9	15	5	9	8	14	9	9	(17)
(18)		PrecStd	79	14	9	14	14	20	28	25	27	20	18	12	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.9	1.1	6.5	21.7	29.4	30.6	31.0	29.5	25.6	18.0	6.6	1.2	(19)
(20)			MCWB	-1.8	-0.4	2.7	11.9	17.2	19.0	21.8	19.7	15.7	10.4	4.6	0.6	(20)
(21)		2%	DB	-2.8	-0.7	4.4	17.4	26.4	28.7	28.8	27.1	22.4	14.9	4.3	-0.2	(21)
(22)			MCWB	-3.5	-1.8	1.3	10.0	15.1	18.9	19.9	18.6	14.5	8.9	3.1	-0.9	(22)
(23)		5%	DB	-4.8	-2.6	2.7	14.3	23.9	26.9	27.3	25.1	19.4	11.9	2.3	-2.2	(23)
(24)			MCWB	-5.4	-3.5	0.6	8.4	14.0	18.6	19.2	17.9	13.2	7.5	1.3	-2.9	(24)
(25)		10%	DB	-7.0	-4.9	1.1	11.4	20.9	25.0	25.5	22.8	16.9	9.4	0.7	-4.4	(25)
(26)			MCWB	-7.5	-5.8	-0.5	6.6	12.7	17.7	18.6	16.8	11.8	6.5	-0.1	-5.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.5	-0.3	3.2	13.0	18.6	22.7	23.0	20.8	17.0	11.2	4.7	0.8	(27)
(28)			MCDWB	-0.9	0.6	5.8	18.8	26.8	27.3	28.2	26.6	23.5	16.3	5.8	1.0	(28)
(29)		2%	WB	-3.6	-1.8	1.8	10.7	16.2	20.8	21.3	19.6	15.1	9.5	3.3	-0.9	(29)
(30)			MCDWB	-2.8	-0.8	3.7	16.5	24.6	26.1	27.0	25.4	20.4	13.7	4.4	-0.2	(30)
(31)		5%	WB	-5.4	-3.4	0.8	8.8	14.6	19.4	20.2	18.6	13.7	8.0	1.4	-2.9	(31)
(32)			MCDWB	-4.9	-2.6	2.5	13.8	22.2	25.0	25.6	23.5	18.7	11.2	2.0	-2.2	(32)
(33)		10%	WB	-7.6	-5.7	-0.4	6.9	13.2	18.2	19.2	17.3	12.3	6.5	0.2	-5.0	(33)
(34)			MCDWB	-7.0	-5.0	1.2	11.1	19.9	23.6	24.0	22.0	16.3	9.3	0.7	-4.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.5	9.6	9.8	9.7	12.2	11.4	10.5	9.8	7.4	6.6	7.9	(35)	
(36)			MCDBR	7.9	7.4	9.3	13.9	16.7	14.9	13.4	14.5	14.7	12.2	5.8	7.4	(36)
(37)			MCWBR	7.7	6.9	7.1	8.3	7.8	7.1	6.3	7.0	7.6	7.2	5.0	7.2	(37)
(38)		5% WB	MCDWB	7.8	7.2	8.2	13.0	15.4	12.8	12.0	12.7	13.3	11.0	5.5	7.3	(38)
(39)			MCWBR	7.7	6.9	6.5	8.1	8.4	6.9	6.2	6.8	7.4	7.0	4.9	7.3	(39)
(40)			Clear-Sky Solar Irradiance	taub	0.277	0.311	0.368	0.408	0.400	0.406	0.428	0.414	0.364	0.349	0.309	0.280
(41)	taud	2.029		2.015	1.928	2.062	2.238	2.297	2.231	2.247	2.346	2.230	2.074	1.971	(41)	
(42)	Ebn at Noon	574		711	758	797	836	836	805	789	781	672	522	442	(42)	
(43)	Edh at Noon	63		97	138	141	126	121	127	117	92	79	60	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.79	1.82	3.35	4.53	5.17	5.78	5.29	4.11	2.82	1.38	0.64	0.48	(44)	
(45)		RadStd	0.07	0.12	0.21	0.39	0.48	0.57	0.33	0.44	0.28	0.13	0.12	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page